Household Vehicle Transactions During and After the Pandemic – A Panel Data Analysis

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Abstract

This paper investigates the factors affecting household vehicle transactions, including additions, removals, and replacements, in the past (spring 2020 to fall 2023) and in the anticipated future (fall 2023 to fall 2026). Using a two-wave panel dataset (n=1,612) from surveys conducted in the U.S., we explore a broad set of hypotheses concerning vehicle transaction dynamics over time. An integrated choice and latent variable model identifies the effects of latent attitudes, sociodemographic characteristics, life events, work arrangements, and COVID-related health concerns in shaping vehicle transaction decisions. Results show that *novelty-seeking* individuals were more likely to engage in future vehicle transactions, whether by increasing, decreasing, or replacing vehicles. Younger adults, households with children, and those experiencing an increase in the number of children or adults showed a higher likelihood of acquiring vehicles, likely in response to an evolving increase in the need for travel. Transitions into the workforce and rising household income further increased the likelihood of vehicle acquisition. An increase in commute frequency reduced the likelihood of vehicle shedding during the pandemic and also increased the likelihood of post-pandemic vehicle acquisition, likely due to a rebound in demand for non-commuting trips. COVID-related health concerns discouraged vehicle shedding during the pandemic. Importantly, past vehicle transactions strongly predict future behaviors. Households that added vehicles during the pandemic were likely to either increase, decrease, or replace vehicles again. In contrast, those that shed vehicles tended to reacquire new vehicles, and those who replaced vehicles were more inclined to do so again in the future.

Keywords: vehicle transaction, COVID-19 pandemic, panel data, integrated choice and latent variable model, life event, work arrangement

1 Introduction

The COVID-19 pandemic substantially disrupted people's activity-travel patterns. During the initial phase of the pandemic, countries around the world witnessed a dramatic reduction in travel by air, mass transit, and other shared modes due to the fear of virus transmission (Shamshiripour et al., 2020). Moreover, a number of studies highlighted a growing reliance on private vehicles (Krolikowski & Naggert, 2021) due to a large proportion of the population perceiving cars as the safest option for their daily travel (Palm et al., 2024; Vega-Gonzalo et al., 2023; Wang et al., 2024; Zarabi et al., 2024). Other studies reported that some non-car owners were compelled to buy or lease one to avoid using mass transportation (Palm et al., 2024; Vega-Gonzalo et al., 2023). At the same time, the automotive industry struggled to meet the increased demand because of a substantial decline in vehicle production and delivery during the pandemic due to business shutdowns, international supply chain disruptions, and shortages of raw materials (Krolikowski & Naggert, 2021). This resulted in a decline in new automotive sales in the United States (Marina et al., 2022) and a boom in used car sales during the pandemic (Rosenbaum, 2020). However, market uncertainty and the overall economic downturn may have also spared some consumer groups from unnecessary hassles and expenses associated with purchasing or replacing cars (de Palma et al., 2022).

Policymakers, auto manufacturers, and related businesses all seek to understand the ways that the pandemic has affected consumer behavior regarding vehicle transactions (i.e., acquisition, disposal, or replacement). However, critical gaps exist in our knowledge about these topics:

- First, although vehicle transaction behavior has been well-studied over the decades prior to the pandemic, research focusing on the dynamics of this major disruption and its potential impact on future patterns in the U.S. has been limited.
- Second, given the relative stability of vehicle market structure prior to the pandemic, many studies have often employed cross-sectional data (Klein & Smart, 2017) or pseudo panel data (Anowar et al., 2016) that cannot include a direct examination of how changes affect the same households over time, and therefore lack statistical power for identifying potentially important dynamic factors. For example, those types of data are limited in their ability to capture the impact of life events (e.g., relocating to a new area, starting or leaving a job) that are critical to understanding the underlying reasons behind decisions in vehicle transaction (Clark et al., 2016; Li, 2024).
- Third, to the best of the authors' knowledge, no prior study has modeled how vehicle transactions in the past might directly impact future transaction decisions under conditions like those experienced during a pandemic (so-called "state dependence"). For instance, consider a household that reduced its travel during the pandemic and consequently shed a vehicle. As travel demand rebounds after the pandemic, many such households may choose to reacquire a vehicle. In this case, the past vehicle disposal itself influences future acquisition decisions by shaping expectations, habits, or perceived constraints, representing true state dependence. Note, however, that another important concept in dynamic modeling is "heterogeneity", which refers to the unobservable differences across individuals that are stable over time, but are not influenced by prior choices. For instance, households that are more novelty-seeking and more car-oriented may be inherently more likely to engage in vehicle transactions, regardless of their past behavior. In this case, observed correlations between past and future transactions reflect stable underlying preferences, rather than causal effect of prior decisions. As suggested

by previous papers (Kitamura & Bunch, 1990), in reality, vehicle transaction decisions often involve both state dependence and heterogeneity, and omitting one effect (e.g., heterogeneity) can result in falsely concluding the existence of the other (in this case, e.g., so-called "spurious state dependence"). The modeling framework proposed here is intended to help properly account for heterogeneity by explicitly incorporating individual-level attitudes.

This study analyzes a two-wave panel dataset collected in the U.S. to investigate households' vehicle transaction-related decisions over two time periods: *actual transaction choices in the recent past* from March 2020 (i.e., at the start of the pandemic) to July/August 2023, and *expected transaction choices during the next three years* from July/August 2023 (hereafter 'fall 2023') to July/August 2026 (hereafter 'fall 2026'). We examine transaction choices defined by the following outcomes (relative to a base option of *no transaction*): (1) an *increase* in the number of vehicles, (2) a *decrease* in the number of vehicles, and (3) no change in vehicle ownership level, but with one (or more) vehicles *replaced.* Using an integrated choice and latent variable (ICLV) model, we identify factors affecting these choices, including: sociodemographic and economic (SDE) characteristics, life events (e.g., starting a job), work arrangement (e.g., adopting a remote work schedule), individual attitudes (e.g., tech-savviness), and COVID-related health concerns.

This paper is structured as follows. Section 2 reviews prior research on the topic. Section 3 describes the data and provides descriptive analyses. Section 4 presents the model framework, Section 5 highlights the model results, and Section 6 discusses findings from the study. Finally, Section 7 concludes with policy implications and future research.

2 Literature Review

Understanding the determinants of vehicle ownership, vehicle holding, and vehicle transaction decisions has long been central to transportation and land use planning. "*Vehicle ownership*" typically refers to whether a household has access to at least one working passenger car, van, pickup, or panel truck, either owned or leased/rented for one month or more and that is kept at the residence and available for the use of household members (Social Explorer, 2018). This concept may be treated as a binary status (yes/no) or specified in terms of the number of vehicles, often referred as "vehicle ownership level" or "vehicle count." *"Vehicle holdings"* generally describes a *static snapshot* of the household vehicle fleet at a given point in time, usually capturing detailed attributes such as the number, type (e.g., sedan, SUV), and ownership status (e.g., owned, leased) of vehicles (Bhat et al., 2009). Additionally, "*vehicle transactions*" refers to *dynamic events* involving the acquisition, disposal, or replacement of vehicles through purchases, sales, leases, or trade-ins. Some studies use "*changes* in vehicle ownership/holding" and "vehicle transaction" interchangeably (Gu et al., 2019; Prevedouros & Schofer, 1992), although the former reflects net changes in the fleet count over time, which may result from one or multiple transactions.

Since our study focuses on *vehicle transactions* (i.e., increase/decrease/replace), the review emphasizes studies examining vehicle ownership level and transaction activities, rather than those centered on vehicle type. Moreover, during our review we found that the definitions of these terms could vary across studies depending on context, and many studies do not clearly

define these terms at the outset. In the following review, we ensure to clarify which aspects of vehicle-related choices individual studies focused on.

2.1 Research Before the Pandemic

Scholars have extensively investigated this topic prior to the pandemic. Some studies analyzed vehicle ownership *at a given timepoint*. SDE characteristics (e.g., work status, household income) and spatial attributes have been found to be associated with vehicle ownership level (Macfarlane et al., 2015; Nobil et al., 1997; Prevedouros & Schofer, 1992). Lifestyle, attitudes, and psychological factors also play significant roles. The *symbolic utility* of a vehicle is positively associated with the stated intention to own a car among current non-vehicle owners (Wu et al., 1999). Households with *pro-car* attitudes tend to own more vehicles, while those with *pro-environmental* attitudes typically own fewer (Flamm, 2009; Lavieri et al., 2017). Furthermore, the adoption of online shopping and remote work was found to be associated with fewer household vehicles (Blumenberg et al., 2021).

Most studies examining *changes in vehicle ownership level over time* rely on cross-sectional data. Lavieri et al. (2017) highlighted that multi-car households tend to make vehicle transactions more often. The presence of children is associated with an increase in vehicle ownership level. Life events, such as changes in household composition, employment, or residential location, significantly influence vehicle ownership level (Scheiner & Holz-Rau, 2013). Individuals with greater loss aversion tend to experience a greater psychological resistance to giving up vehicles (Grush & Niles, 2018). In addition, changes in vehicle ownership level are closely tied to accessibility of alternative travel options. For instance, access to affordable and efficient public transportation systems, along with shared mobility services such as carsharing, ridehailing, and micromobility, are positively associated with having fewer vehicles per adult and with an increased the likelihood for being a zero-vehicle household (Blumenberg et al., 2021; Cullinane, 2002; Liu & Cirillo, 2015).

It has become widely recognized that behavioral models based solely on cross-sectional data have inherent limitations (Dougherty, 2006; Kitamura & Bunch, 1990). However, dynamic panel models of vehicle ownership remain limited due to the scarcity of panel datasets. Using the four-wave British Household Panel Survey (1993-1996), Hanly & Dargay (2000) applied a latent regression model to demonstrate that current vehicle ownership levels are impacted by the past ownership, suggesting the effects of habit persistence or loyalty (i.e., the "Feedback Effect"). Yamamoto (2008) estimated a duration model on a 15-wave household panel dataset, showing that life-course events act as catalysts for both acquisition and disposal of vehicles, resulting in changes to vehicle ownership level. Similarly, Goodwill (1993), using UK panel data, found that income, age, previous vehicle ownership, and life-course events between survey waves are associated with changes in vehicle ownership level. Hanly & Dargay (2000) as well as Fatmi & Habib (2016) further highlighted that the effects of life events vary substantially across population segments, with changes in vehicle ownership level often taking several years to materialize – a pattern consistent with durable goods acquisition behaviors (Mannering & Winston, 1985).

2.2 Research During and After the Pandemic

The COVID-19 pandemic has significantly reshaped many aspects of individuals' lives, including work arrangements (Iogansen et al., 2024), time use (Batur et al., 2023), travel

behavior (Wang et al., 2024; Zarabi et al., 2024), and participation in in-person and online activities (Chowdhury et al., 2024; Matson et al., 2022). While some of these impacts were temporary, many are expected to have long-lasting effects.

The pandemic likely deterred some consumers from purchasing a vehicle due to some financial factors such as income and wealth losses (Coibion et al., 2020). Rising car prices and housing expenses during the pandemic have also been reported to negatively affect vehicle acquisition among U.S. residents (Thakuriah, 2024). The same study shows that, conversely, government-issued economic stimulus funds increased their odds of obtaining a vehicle.

During the pandemic, individuals appeared to place greater value on the convenience of private cars, particularly the ability to travel spontaneously whenever and wherever needed (Moody et al., 2021). The perceived safety of private cars also became more important compared to the pre-pandemic period (Wang et al., 2024). In line with these findings, Palm et al. (2024) found that individuals who had contracted COVID-19 or lived with someone who had were more likely to purchase a vehicle.

Studies also indicate that the vehicle ownership, usage, and purchase intentions during and after the pandemic varied across different population groups. The pandemic appeared to induce more car use and to increase the intention to purchase vehicles, particularly among individuals who were less car dependent prior to the pandemic, such as younger individuals and frequent transit riders (Palm et al., 2024; Vega-Gonzalo et al., 2023). Conversely, minorities, low-skilled workers, and residents of multi-family dwellings were less likely to increase their vehicle ownership level. High-income, well-educated teleworkers were found to temporarily reduce their car use to a larger extent, while low-income individuals tend to maintain similar levels of car use.

While previous studies provide valuable insights into how SDE factors and health concerns impacted vehicle-related choices, none have analyzed longitudinal data to explore vehicle transactions across multiple timepoints. Additionally, few studies have incorporated attitudes and life events into their analyses, both of which are critical for understanding the underlying motivations towards vehicle transaction decisions. This study seeks to address these gaps.

3 Data and Variables

3.1 COVID Mobility Survey Datasets

This study uses survey data collected through the COVID-19 Mobility Study, which aims to provide insights into the short-term and long-term impacts of the pandemic on consumer behavior and mobility decisions in the U.S. To achieve this, the team designed and distributed four surveys with a rotating panel structure during the pandemic, with the first survey conducted in spring 2020 (May-August 2020, n=13,658), the second in winter 2020 (December 2020-January 2021, n=7,983), the third in fall 2021 (August-October 2021, n=14,084) and the fourth in fall 2023 (September-December 2023, n=6,469). At the time of writing, an additional wave of data collection is planned for Spring 2025. The mobility panel study also benefits from previous rounds of data collection conducted before the pandemic, in 2015, 2018 and 2019 in California and other selected regions of the U.S., with some respondents participating in multiple waves. Additional details about the project and the various waves of data collection is available on the project website (https://postcovid19mobility.ucdavis.edu), as well as in existing project reports

(Circella et al., 2023; Ozbilen et al., 2024) and published journal papers (Compostella et al., 2023; Iogansen et al., 2024; Matson et al., 2023).

For all surveys, the team used a combination of sampling and recruitment approaches to reach a broader and more diverse pool of participants as well as to reduce selection bias. These approaches included: (1) recontacting survey-takers from the previous survey waves in the project; (2) recruiting new participants via an opinion panel with a quota sampling method; (3) recruiting new participants via convenience sampling through posts on social media, professional email lists, local listservs, and personal connections; and (4) in fall 2021 and fall 2023, also inviting California residents via stratified random sampling and having them participated in the study either with a printed questionnaire or through an online survey. A mix of these approaches allowed us to offset known shortcomings and recruit segments of population who are challenging to reach.

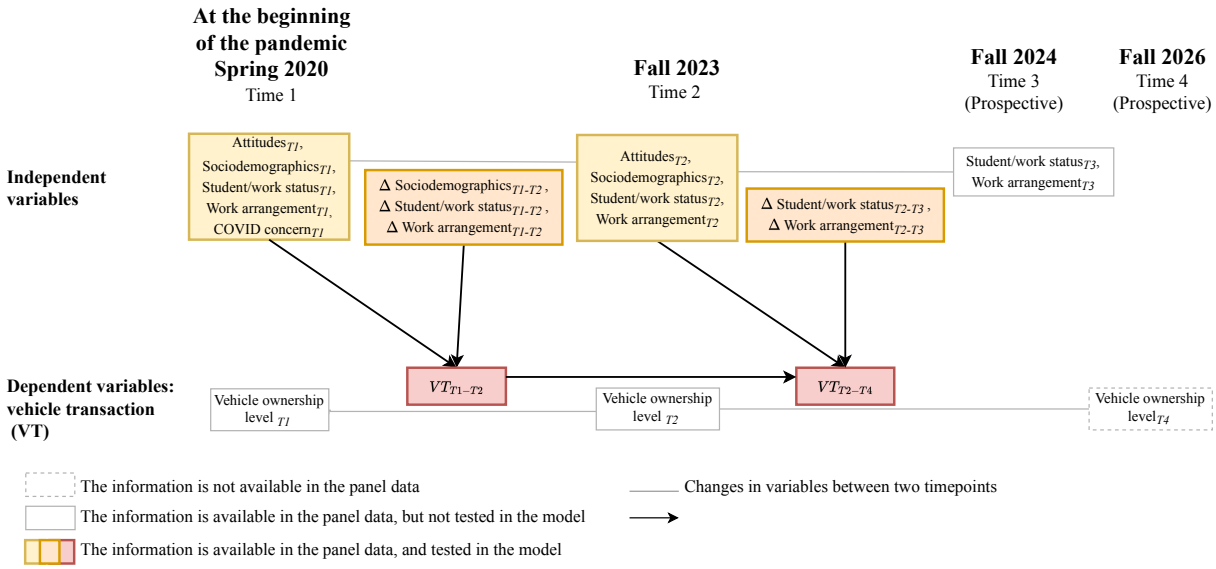
Each wave collected detailed information on respondents' SDE traits, technology use, online shopping habits, transportation mode use, vehicle ownership level, and major life events that occurred prior to and during the pandemic. The surveys also included a series of attitudinal statements on various topics such as environment concerns, lifestyle choices, and car usage. To better capture changes during and after the pandemic, some questions in the surveys elicited responses for both the present (i.e., at the time of data collection), the past (via retrospective recall), and the future (via expectation) situations.

The present study focuses on a cohort of 1,612 longitudinal respondents who participated in both the spring 2020 and fall 2023 surveys in the U.S. Compared to the U.S. population[1], this longitudinal sample has a lower representation of younger individuals, males, non-White and Hispanic individuals, and those with lower income levels. Consistent with findings in the existing literature on panel studies, these survey participants are either more challenging to reach initially (Stempowski, 2023) or generally have lower retention rates for subsequent surveys (Satherley et al., 2015; Van Wissen & Meurs, 1989; Wang et al., 2023). Note that, while a representative sample may be desirable, the focus of our study is not on obtaining population-level estimates or forecasts related to vehicle transactions, for which probability-based sampling and subsequent weighting are appropriate. Instead, our focus is on identifying factors behind vehicle transactions via regression analysis with various explanatory variables, an approach for which representativeness and weighting are less critical.

3.2 Dependent and Independent Variables

Figure 1 presents a set of key variables, collected from the spring 2020 and fall 2023 surveys and grouped by several reference timepoints. Self-reported *household vehicle transactions in two time periods* are the choice outcomes of interest. Factors impacting these choices include attitudes, SDE characteristics, as well as changes in SDE characteristics and life events between the two timepoints. The models presented in Section 4 analyze the color-coded variables shown in this figure.

[1] For detailed comparisons between the sample distribution and the target distribution in the U.S. population, refer to Table 1 in the Supplemental Material.



Note: The variables VT_{T1-T2} and VT_{T2-T4} are derived directly from two survey questions in the fall 2023 survey and validated by cross-checking against actual household vehicle ownership levels at three timepoints (i.e., $Vehicle\ ownership\ level\ _{T1,T2,T4}$) collected through the spring 2020 and fall 2023 surveys.

Figure 1 Key variable groups by timepoint

This study focuses on two dependent variables: (1) the *past vehicle transaction choice during the period from the beginning of the pandemic (spring 2020) to fall 2023*, and (2) the *expected vehicle transaction choice over the next three years (fall 2023 to fall 2026)*. The two dependent variables were obtained from the fall 2023 dataset and validated using information about actual household vehicle ownership levels collected in the spring 2020 and fall 2023 surveys. As illustrated in Figure 2, between the start of the pandemic and fall 2023, 7.3%, 7.4% and 23.6% of respondents reported adding, shedding, or replacing their vehicles, respectively. Looking three years ahead, 7.8%, 4.2% and 32.8% of respondents expected to increase, decrease, or replace their household fleets, respectively. Notably, the proportion of individuals replacing vehicles has been, and is projected to remain, higher than those adding or shedding vehicles. While past trends show a nearly equal proportion of individuals reporting they had increased or decreased their number of vehicles in the household, future expectations reveal a larger share of individuals planning to increase (7.8%) rather than decrease (4.2%) their household vehicle ownership level.

We hypothesize that changes in vehicle ownership level may be directly impacted by household dynamics, such as individuals moving in or out of the household and bringing or taking their vehicles with them. To account for this potentially confounding factor, Figure 3 presents the same trends as Figure 2 but focuses only on households that maintained the same composition (i.e., the same number of adults and children) during the study period. As expected, the proportion of individuals who altered their vehicle ownership level (including increases, decreases and replacements) in Figure 3 is lower than Figure 2.

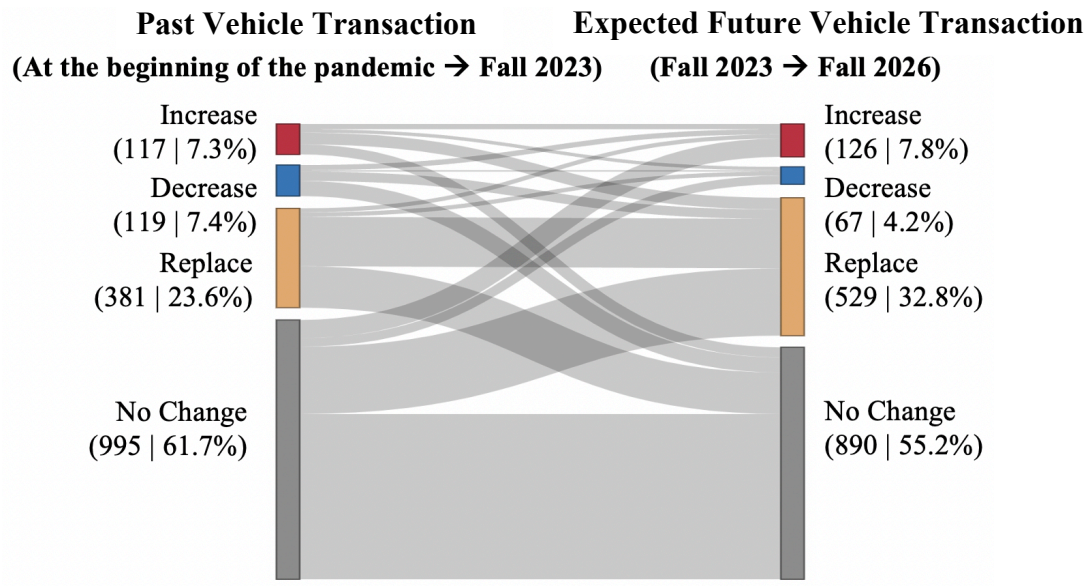


Figure 2 Past and expected future household vehicle transactions (n=1,612)

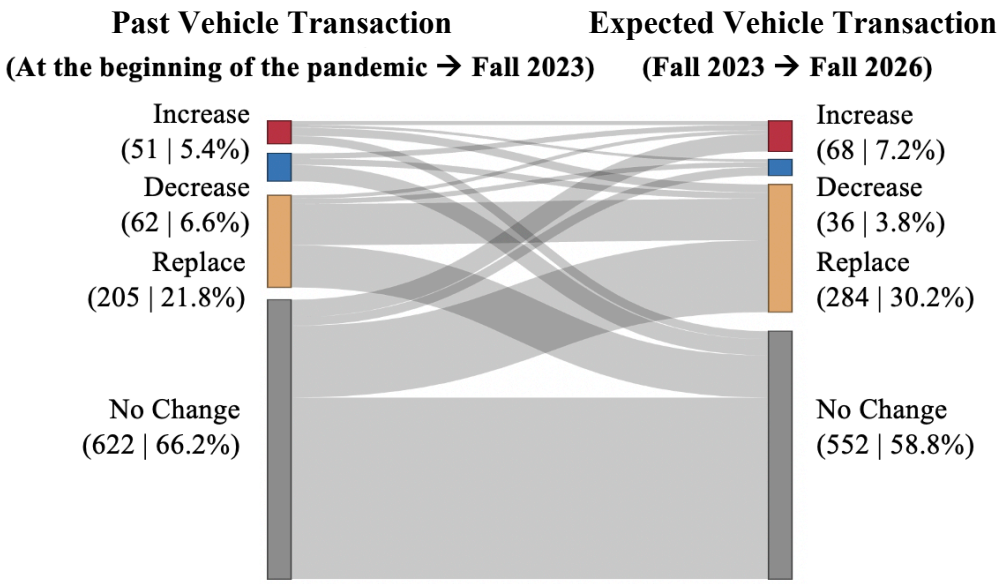


Figure 3 Past and expected future household vehicle transactions, for households that maintained the same composition (n=940)

We now discuss the variables used for modeling the recent and expected future transactions. Beyond standard SDE characteristics (e.g., birth cohort, gender, ethnicity, race), we explore a range of additional factors, including life events, work arrangements, general attitudes, and COVID-related health concerns. Table 1 and Table 2 presents the summary statistics of all variables tested in the model. Note that some variables are available for only one timepoint. More specifically, Table 1 summarizes various characteristics at one timepoint, reflecting baseline household needs and longer-term constraints, while Table 2 summarizes life events and changes in work arrangement, capturing triggering events that may prompt vehicle transactions during the study period. Diagnostic checks, including variance inflation factors and pairwise

correlations, indicate that these variables do not exhibit problematic multicollinearity and can be jointly interpreted within the model.

We measured general attitudes by analyzing responses to 18 attitudinal statements (rated on a five-point Likert-type scale from "strongly disagree" to "strongly agree"), collected at two timepoints. Using two exploratory factor analyses (EFA) with "promax" rotation[2], these statements were distilled into six underlying psychological constructs. The *pro-environment* factor reflects support for environmental regulations that raise the cost of driving to mitigate the negative impacts of transportation on the environment and fund better public transportation. The *pro-driving* factor captures a preference towards owning a vehicle and the enjoyment of driving. The *pro-active* factor highlights the value placed on active lifestyle through regular walking and exercise. The *novelty-seeking* factor indicates familiarity with and interest in new technologies and new experiences. The *car-captive* factor represents dependence on vehicles due to limited access to alternative travel modes and inconvenient schedules. Finally, the *pro-urban* factor reflects a preference for neighborhoods prioritizing urban amenities over spacious housing. Note that, COVID-related health concerns are measured by asking respondents for the best response to a single statement on a five-point ordinal scale ranging from "not concerned" to "strongly concerned". As such, we estimate its effects as an independent variable in the choice model, instead of having it included in the measurement model.

We examined various life events and their potential impact on vehicle transactions. Changes in educational attainment and student/work status may influence activity/travel patterns related to commuting, which accounted for 15% of daily trips (before the pandemic), with 91% of workers using personal vehicles (Bureau of Transportation Statistics, 2017). From spring 2020 to fall 2023, 12.0% of respondents transitioned from being non-workers to workers, reflecting employment recovery after pandemic-related job losses. Similarly, 27.9% reported income increases compared to 11.7% reporting declines. Further, as of fall 2023, more respondents anticipated starting or resuming work in a year (i.e., by fall 2024) than those planning to leave the workforce. Changes in household composition, such as shifts in the number of children (under 18), adults (18-64), or older adults (65 or over), may also impact vehicle needs. Studies have demonstrated that households often add or replace vehicles to accommodate changing needs associated with children (Lavieri et al., 2017). Finally, residential relocation, particularly when accompanied by significant changes in built environment characteristics, is expected to affect vehicle reliance. For instance, moving from rural to urban areas may lead to shedding vehicles due to better access to jobs and destinations. The adoption of hybrid and remote work arrangements likely impacts commuting frequency and vehicle needs. Notably, between spring 2020 and fall 2023, respondents in this study reduced monthly commuting days by an average of 2.1 and increased monthly remote working days by an average of 3.4. Finally, we hypothesize that individuals with greater COVID-related health concerns may be more inclined to increase or maintain their vehicle ownership level, driven by fears of shared travel modes.

After careful consideration, we chose to not include certain variables in the model. For instance, past vehicle ownership level is highly correlated with recent changes, making it a strong explanatory variable. However, vehicle count in the past, as an outcome of various other factors, may absorb much of the explanatory power and statistical significance, limiting the insights into how other factors influence current and future vehicle count. Additionally, trip frequency and mode use are closely related to changes in vehicle count, but their bidirectional

[2] For more detailed overview of the factor analysis results, please refer to Table 2 in the Supplemental Material.

relationship could introduce endogeneity (Goodwin & Mogridge, 1981) if included as explanatory variables.

Table 1 Past and expected future vehicle transactions by various characteristics at one timepoint (N=1,612)

				Past Vehicle Transaction (spring 2020 to fall 2023)						Expected Future Vehicle Transaction (Fall 2023 to Fall 2026)			
Variables	**Categories**	**n**	**% / mean**	**Increased 117 7.3%**	**Decreased 119 7.4%**	**Replaced 381 23.6%**	**No transaction 995 61.7%**	**n**	**% / mean**	**Increase 126 7.8%**	**Decrease 67 4.2%**	**Replace 529 32.8%**	**No transaction 890 55.2%**
General Attitudes[1]													
Pro-environment			0.00	-0.05	-0.08	-0.10	**0.05**[5]		0.00	0.06	-0.13	**0.20**	-0.12
Pro-driving			0.00	**0.20**	-0.01	0.18	-0.09		0.00	-0.03	**0.11**	-0.03	0.01
Novelty-seeking			0.00	**0.47**	-0.23	0.16	-0.09		0.00	**0.35**	-0.03	0.19	-0.16
Pro-active			0.00	0.03	-0.20	**0.07**	-0.01		0.00	-0.13	0.07	**0.17**	-0.09
Car-captive			0.00	0.17	0.19	**0.26**	-0.12		0.00	-0.12	0.11	**0.19**	-0.10
Pro-urban			0.00	-0.21	-0.24	-0.11	**0.10**		0.00	-0.09	0.01	-0.07	**0.06**
SDE Characteristics													
Birth cohort[2,3]	Millennials or younger (after 1980)	397	24.6%	22.0%	**40.2%**	21.0%	27.8%	397	24.6%	20.9%	**45.2%**	29.9%	25.3%
	Generation X (1965-1980)	478	29.7%	27.7%	**44.4%**	22.7%	32.3%	478	29.7%	28.3%	**32.5%**	26.9%	31.6%
	Baby Boomers or older (before 1965)	737	45.7%	50.3%	15.4%	**56.3%**	39.9%	737	45.7%	**50.8%**	22.2%	43.3%	43.1%
Ethnicity[3]	Non-Hispanic/Latino/Spanish origin	1465	90.9%	88.0%	**93.3%**	90.0%	91.3%	1465	90.9%	84.1%	89.6%	90.4%	**92.2%**
	Hispanic/Latino/Spanish origin	147	9.1%	**12.0%**	6.7%	10.0%	8.7%	147	9.1%	**15.9%**	10.4%	9.6%	7.8%
Race[3]	Non-white	371	23.0%	21.4%	16.8%	22.0%	**24.3%**	371	23.0%	**36.5%**	13.4%	23.4%	21.6%
	White	1241	77.0%	78.6%	**83.2%**	78.0%	75.7%	1241	77.0%	63.5%	**86.6%**	76.6%	78.4%
Gender[3]	Not-Female	629	39.0%	**47.0%**	33.6%	38.1%	39.0%	629	39.0%	33.3%	40.3%	**45.0%**	36.1%
	Female	983	61.0%	53.0%	**66.4%**	61.9%	61.0%	983	61.0%	**66.7%**	59.7%	55.0%	63.9%
Educational attainment	Lower than bachelors	526	32.6%	31.6%	33.6%	27.0%	**34.8%**	492	30.5%	34.1%	20.9%	22.7%	**35.4%**
	Bachelor or higher	1086	67.4%	68.4%	66.4%	**73.0%**	65.2%	1120	69.5%	65.9%	**79.1%**	77.3%	64.6%
Student status	Not a student	1517	94.1%	90.6%	**95.8%**	93.4%	94.6%	1548	96.0%	88.9%	95.5%	**96.8%**	96.6%
	Student	95	5.9%	**9.4%**	4.2%	6.6%	5.4%	64	4.0%	**11.1%**	4.5%	3.2%	3.4%
Work status	Non-workers	733	45.5%	29.9%	**57.1%**	38.3%	48.6%	621	38.5%	25.4%	26.9%	31.4%	**45.4%**
	Workers	879	54.5%	**70.1%**	42.9%	61.7%	51.4%	991	61.5%	**74.6%**	73.1%	68.6%	54.6%
Possess of a driver's license	No	53	3.3%	1.7%	1.7%	0.8%	**4.7%**	53	3.3%	**7.1%**	3.0%	0.8%	4.4%
	Yes	1559	96.7%	98.3%	98.3%	**99.2%**	95.3%	1559	96.7%	92.9%	97.0%	**99.2%**	95.6%
Household income	<=49,999	390	24.2%	13.7%	21.8%	18.6%	**27.8%**	343	21.3%	**27.8%**	13.4%	9.8%	**27.8%**
	50,000-99,999	530	32.9%	34.2%	**34.5%**	29.7%	33.8%	514	31.9%	29.4%	29.9%	**32.5%**	32.1%
	100,000 or over	692	42.9%	**52.1%**	43.7%	51.7%	38.4%	754	46.8%	42.9%	56.7%	**57.7%**	40.1%
Household size	One member	340	21.1%	7.7%	10.1%	11.3%	**27.7%**	376	23.3%	12.7%	16.4%	12.5%	**31.7%**
	Two members	637	39.5%	26.5%	**52.1%**	39.9%	39.3%	679	42.1%	27.0%	**47.8%**	47.6%	40.6%
	Three or more members	637	39.5%	**65.8%**	37.8%	48.8%	33.0%	558	34.6%	**60.3%**	35.8%	39.9%	27.8%
Presence of children	No	1198	74.3%	42.7%	77.3%	65.6%	**80.9%**	1251	77.6%	49.2%	82.1%	74.3%	**83.3%**
	Yes	414	25.7%	**57.3%**	22.7%	34.4%	19.1%	361	22.4%	**50.8%**	17.9%	25.7%	16.7%
Housing tenure	Rent or other	571	35.4%	27.4%	32.8%	29.4%	**38.9%**	482	29.9%	**41.3%**	23.9%	23.3%	32.7%
	Own	1041	64.6%	**72.6%**	67.2%	70.6%	61.1%	1130	70.1%	58.7%	76.1%	**76.7%**	67.3%
Neighborhood type	Urban	532	33.0%	29.9%	26.9%	26.0%	**36.8%**	508	31.5%	**42.1%**	32.8%	23.4%	34.6%
	Suburban	914	56.7%	**63.2%**	55.5%	63.0%	53.7%	929	57.6%	45.2%	55.2%	**65.8%**	54.7%
	Rural	166	10.3%	6.8%	**17.6%**	11.0%	9.5%	176	10.9%	**12.7%**	11.9%	10.8%	10.7%
Work Arrangement before the Pandemic (fall 2019, retrospectively reported in spring 2020) **vs. fall 2023**													
Monthly commuting days			10.0	**13.0**	8.2	11.2	9.4		7.9	**10.1**	8.8	9.0	6.8
Monthly remote working days			2.2	**2.8**	2.3	2.1	2.2		5.6	7.2	**7.6**	5.6	5.3
COVID-related Health Concerns (only measured in spring 2020)													
Level of concerns	Not concerned or neutral	123	7.6%	10.3%	**13.4%**	7.3%	6.7%	---[4]	---	---	---	---	---
	Somewhat concerned	442	27.4%	**29.9%**	24.4%	28.9%	26.8%	---	---	---	---	---	---
	Strongly concerned	1048	65.0%	59.8%	62.2%	63.8%	**66.4%**	---	---	---	---	---	---

Notes:

[1] Numbers denote the average factor scores by the category of vehicle transaction.
[2] A birth cohort refers to a group of individuals who were born around the same time period from a particular population that typically shares certain events and experiences over their life course.
[3] Birth cohort, ethnicity, race, and gender are considered as static variables, indicating that they typically remain constant for the same individual over time and do not change, with rare exceptions such as transgender individuals. The rest of variables are considered as time-varying or dynamic variables.
[4] Cells with "---" indicate variables that are not tested in the model due to a lack of information.
[5] Bolded values represent the highest value within each respective row, while values with underscore represent the lowest value within each respective row.

Table 2 Past and expected future vehicle transactions by life events and changes in work arrangement between two timepoints (N=1,612)

		Past Vehicle Transaction (spring 2020 to fall 2023)						Expected Future Vehicle Transaction (fall 2023 to fall 2026)					
Variables	**Categories**	**n**	**% / mean**	**Increased 117 7.3%**	**Decreased 119 7.4%**	**Replaced 381 23.6%**	**No transaction 995 61.7%**	**n**	**% / mean**	**Increase 126 7.8%**	**Decrease 67 4.2%**	**Replace 529 32.8%**	**No transaction 890 55.2%**
Life Events and Changes in Work Arrangements													
Changes in education attainment	No change	1510	93.7%	90.6%	**96.6%**	94.5%	93.4%	---[1]	---	---	---	---	---
	Increase	102	6.3%	**9.4%**	3.4%	5.5%	6.6%	---	---	---	---	---	---
Changes in student status	No change	1517	94.1%	90.6%	**95.8%**	92.9%	94.8%	1570	97.4%	92.9%	**98.5%**	96.8%	98.3%
	Student --> Non-student	63	3.9%	**5.1%**	3.4%	5.0%	3.4%	18	1.1%	**4.0%**	1.5%	1.3%	0.6%
	Non-student --> Student	32	2.0%	**4.3%**	0.8%	2.1%	1.8%	24	1.5%	**3.2%**	0.0%	1.9%	1.1%
Changes in work status	No change	1340	83.1%	82.9%	**85.7%**	82.2%	83.1%	1528	94.8%	90.5%	92.5%	95.1%	**95.4%**
	Worker --> Non-worker	81	5.0%	3.4%	5.0%	4.7%	**5.2%**	31	1.9%	1.6%	**6.0%**	2.5%	1.3%
	Non-worker --> Worker	193	12.0%	**13.7%**	9.2%	13.1%	11.7%	53	3.3%	**7.9%**	1.5%	2.5%	3.3%
Changes in household income	No change	974	60.4%	48.7%	59.7%	57.0%	**63.1%**	---	---	---	---	---	---
	Decrease	189	11.7%	7.7%	**15.1%**	11.0%	12.1%	---	---	---	---	---	---
	Increase	450	27.9%	**43.6%**	25.2%	32.0%	24.8%	---	---	---	---	---	---
Changes in the number of children (<18)	No change	1367	84.8%	60.7%	84.0%	81.1%	**89.1%**	---	---	---	---	---	---
	Decrease	156	9.7%	**28.2%**	10.1%	10.5%	7.2%	---	---	---	---	---	---
	Increase	89	5.5%	**11.1%**	5.9%	8.4%	3.6%	---	---	---	---	---	---
Changes in the number of adults (18-64)	No change	1096	68.0%	61.5%	68.1%	65.6%	**69.6%**	---	---	---	---	---	---
	Decrease	372	23.1%	15.4%	23.5%	**24.4%**	23.4%	---	---	---	---	---	---
	Increase	143	8.9%	**23.1%**	8.4%	10.0%	6.9%	---	---	---	---	---	---
Changes in the number of old adults (65+)	No change	1336	82.9%	**90.6%**	78.2%	85.0%	81.7%	---	---	---	---	---	---
	Decrease	61	3.8%	2.6%	**7.6%**	3.7%	3.5%	---	---	---	---	---	---
	Increase	214	13.3%	6.8%	14.3%	11.3%	**14.8%**	---	---	---	---	---	---
Changes in residential relocation	No	1217	75.5%	67.5%	76.5%	74.5%	**76.7%**	---	---	---	---	---	---
	Yes	395	24.5%	**32.5%**	23.5%	25.5%	23.3%	---	---	---	---	---	---
Changes in the level of urbanization	No change	1356	84.1%	82.1%	81.5%	83.5%	**84.9%**	---	---	---	---	---	---
	Decrease	143	8.9%	**13.7%**	10.9%	9.2%	8.0%	---	---	---	---	---	---
	Increase	111	6.9%	4.3%	**7.6%**	7.3%	7.0%	---	---	---	---	---	---
Changes in commuting days			-2.1	-2.5	-3.0	-3.1	**-1.6**		-0.1	-0.5	-0.8	-0.1	**0.1**
Changes in remote working days			3.4	**5.9**	3.6	4.3	2.7		0.03	0.1	0.04	**0.4**	-0.2

Notes:
[1] Cells with "---" indicate variables that are not tested in the model due to a lack of information.
[2] Bolded values represent the highest value within each respective row, while values with underscore represent the lowest value within each respective row.

4 Method

In this study, we use an ICLV model to estimate a two-part dynamic model that jointly determines two dependent variables. Each part combines two sub-models: a latent variable model and a discrete choice model (Abou-Zeid & Ben-Akiva, 2014). The latent variable model addresses the relationships between observable features of the individuals (such as SDE, neighborhood characteristics, COVID-19 health concerns, and underlying psychometric factors). The discrete choice model assumes that (unobservable) utilities associated with the vehicle transaction options determine the outcomes, and that these can be modeled as a function of latent attitudes, individual and household characteristics, and life events. Figure 4 depicts the modeling framework in this study, and Equations (1) through (4) comprise the mathematical representation of the model (Hess et al., 2018).

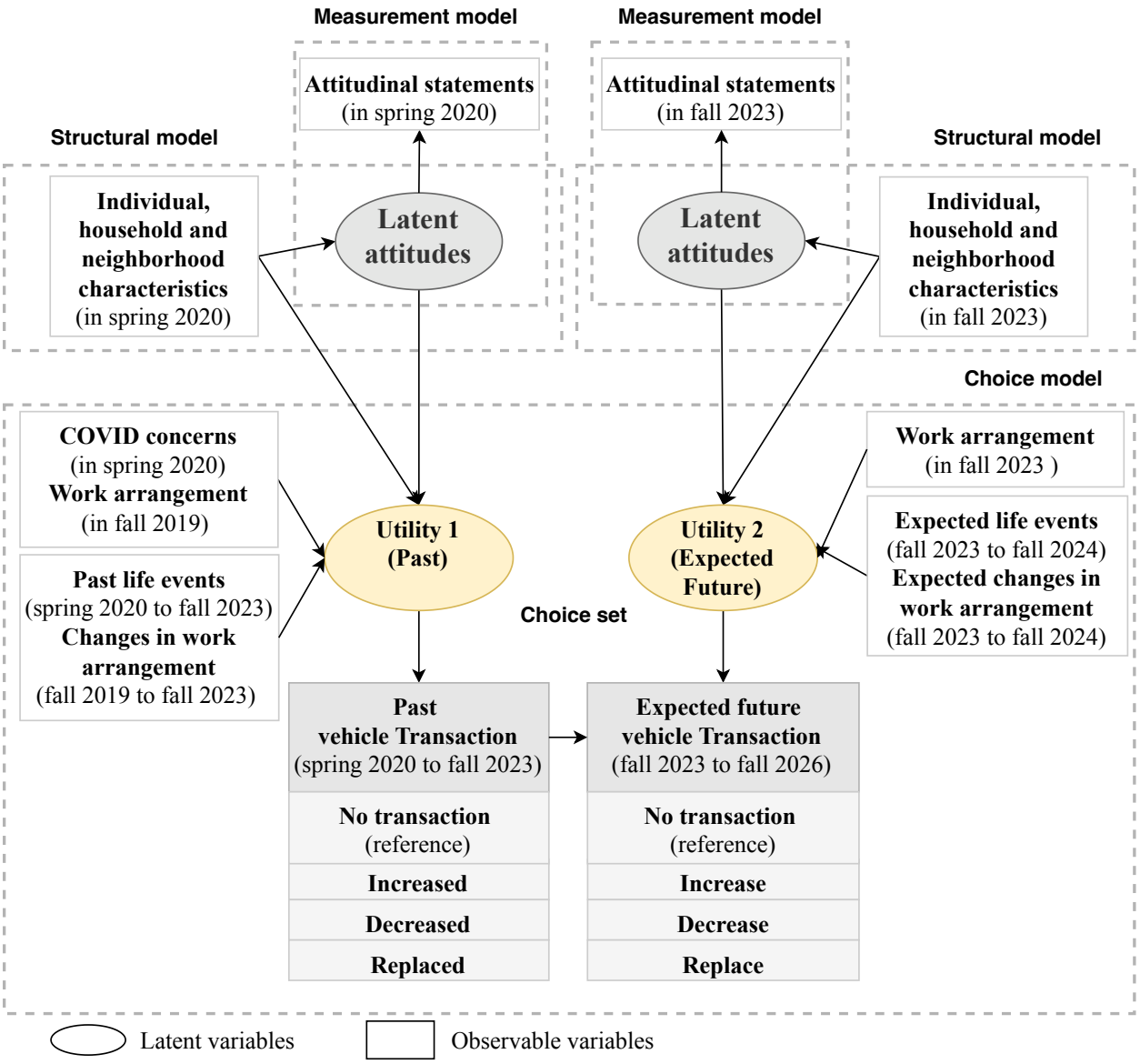


Figure 4 Modeling framework of the ICLV model

In the choice model, two multinomial logit-kernel sub-models (MNL) are used to compute the conditional probabilities for the two transaction outcomes[3] (Abdulsalam et al., 2013). There are four mutually exclusive outcomes j ($j = 1{:}\, no\ change,\ 2{:}\, increase,\ 3{:}\, decrease,\ and\ 4{:}\, replace$) for each choice situation (or corresponding timepoint) m ($m = 1{:}\, past\ transaction, 2{:}\, expected\ future\ transaction$).

Each individual n is assumed to choose the alternative j that yields the highest utility $\mathrm{u_{mnj}}$ in each choice situation (as shown in Eq. (4)). Furthermore, the utility is a function of latent variables x_n^*, individuals' observable features s_n at one timepoint and observable changes in features between two timepoints Δs_n (i.e., life events, changes in work arrangements) (as shown in Eq. (1))[4]. In addition, we expect that transaction in the past had direct impacts on the expected future transaction.

We estimated two latent variable (LV) models separately for the two MNL models described before. Each LV model consists of two components. The first component is the measurement equations, which estimate the impact of latent variables x_n^* on the attitudinal responses i_n (as shown in Eq. (2)). The initial specifications of these measurement equations were informed by the results of the EFA. Since the psychological constructs of individuals at two timepoints are rather similar, we decide to constrain the matrix of factor loadings D to be the same for the two measurement models. The second component of the LV model contains the structural equations, which define each latent variable $\mathrm{x_n^*}$ as a function of individuals' observable features s_n (as shown in Eq. (3)). The model was estimated in R package Apollo (Hess & Palma, 2019) using the Bunch et al. (1993) maximum likelihood estimation algorithm.

Utility specification:

$$\mathrm{u_{mnj}} = \mathrm{B_{mj}}(\mathrm{s_n} + \mathrm{D}s_n) + \Gamma_{\mathrm{mj}}\mathrm{x_n^*} + \epsilon_{\mathrm{mnj}}\ ,\ \epsilon_{m\mathrm{nj}} \sim i.i.d.\, Gumbel \quad (1)$$

$\mathrm{u_{mnj}}$: utility of alternative j in choice situation m for individual n; $\mathrm{s_n}$: observable features (e.g., work status) of individual n, reported at one timepoint; Δs_n: observable changes in features (e.g., life events) between two timepoints; $\mathrm{B_{mj}}$: unknown coefficients between each observable feature and each alternative j in choice situation m to be estimated; $\mathrm{x_n^*}$: latent factors of individual n; Γ_{mj}: unknown coefficients between each latent factor and alternative j in choice situation m; ϵ_{mnj}: random disturbances of unobserved component with *i.i.d.* Gumbel distribution.

Measurement equation:

$$\mathrm{i_{mn}} = \mathrm{D}\mathrm{x_{mn}^*} + \eta_{\mathrm{mn}}\ ,\ \eta_{\mathrm{mn}} \sim N(0, \Psi) \quad (2)$$

[3] The MNL model assumes the independence of irrelevant alternatives (IIA) property, which states that the ratio of the choice probabilities between two alternatives is unaffected by the existence or attributes of other alternatives in the choice set (Cheng & Long, 2007; Vijverberg, 2011). To test if IIA holds true, we turn to a nested logit (NL) model, which places similar alternatives within the same nest and distinct alternatives across distinct nests (McFadden, 1978; Williams, 1977). Note that the Lambda parameter, unique to the nest logit, helps determine whether nested structures explain the choice of interest better than non-nested structures, or MNL. We find lambda parameters from the nest logit models is larger than 1, indicating that the nesting structure is invalid and is insistent with the random utility maximization.

[4] For brevity in this explanation and the equations below, the time variable t is omitted from the notation. However, when modeling recent and expected changes in vehicle count, individual and household characteristics as well as life events reflect specific timepoints or changes between different time periods.

i_{mn}: the level of the agreement to each attitudinal statement of individual n in each timepoint; D: factor loadings indicating the relationship between each indicator and latent factors, and were kept the same for both timepoints[5]; η_{mn}: measurement errors with normal distribution $N(0, \Psi)$.

Structural equation:

$$x^*_{mn} = A_m s_n + v_{mn}\ , v_{mn} \sim N(0, \Phi) \qquad (3)$$

A_m: unknown coefficients between observable features and each latent factor to be estimated for each timepoint; v_n: random disturbances with normal distribution $N(0, \Phi)$; Φ: variance-covariance matrix.

Choice equation:

$$y_{mnj} = \begin{cases} 1, \text{if } u_{mnj} \geq u_{mnj'}, \forall j' \\ 0, \text{otherwise} \end{cases} \qquad (4)$$

y_{mnj}: final choice of individual n among alternative j in choice situation m.

5 Model Results

We here present the results of three sub-models one after another, which were estimated simultaneously. To achieve a more parsimonious model, only variables at a level of significance of 10% were included in the final model specification.

5.1 Individual Attitudes

The measurement model identified four significant latent variables: *pro-environment*, *pro-driving*, *pro-active* and *novelty-seeking*. Table 3 displays the coefficients of the indicators in the measurement equation, where higher absolute values indicate a stronger relationship between the latent variables and the respective indicators.

Table 3 Results of the measurement model (N=1,612)

Latent factors and corresponding attitudinal statements	Coefficient	Robust t-stat[1] and sig.[2]
Pro-environment		
We should raise the cost of driving to provide funding for better public transportation.	1.16	(48.00)***
We should raise the cost of driving to reduce the negative impacts of transportation on the environment.	1.10	(44.29)***
I always think about ways in which I can reduce my impact on the environment.	0.40	(17.03)***
Pro-driving		
I prefer to be a driver rather than a passenger.	0.82	(24.66)***
I like driving a car.	0.80	(20.01)***
I definitely want to own a car.	0.57	(14.31)***
To me, a car is just a way to get from place to place.	-0.49	(-15.14)***

[5] Theoretically, these coefficients could vary over time; however, a prior study (Mirtich et al., 2021) suggests that individual attitudes tend to remain relatively stable across demographic groups. Our own initial factor analysis using data from two timepoints also produced highly similar factor loadings, supporting this assumption. Furthermore, given the already complex structure of our model, this issue is beyond the scope of the current study and could be explored in future research.

Pro-active		
I like riding a bike.	0.39	(7.88)***
I like walking.	0.36	(7.18)***
Getting regular exercise is very important to me.	0.33	(7.29)***
Novelty-seeking		
I like to be among the first people to have the latest technology.	0.69	(21.40)***
I will stretch my budget to buy something new and exciting.	0.54	(17.42)***
Having Wi-Fi and/or good internet access on my mobile phone everywhere I go is essential to me.	0.46	(18.31)***
I like trying things that are new and different.	0.34	(13.29)***

Notes:
[1] Robust t-statistic is less sensitive to violations of assumptions than the traditional t-statistic by using methods that are less affected by outliers or non-normality.
[2] Significance level (*10%, **5%, ***1%)

5.2 Explanatory Variables Associated with Attitudes

Table 4 presents the coefficients of the two structural models for the same individual using observations from spring 2020 and fall 2023, respectively. Overall, the level of significance of variables and the magnitude of their impacts are largely consistent between the two timepoints for each factor. In both timepoints, the *pro-environment* attitude was stronger among individuals with higher education (Dietz et al., 2002) and urban residents (Ambrosius & Gilderbloom, 2015). Members of the youngest birth cohort is found to be more *pro-environment* in spring 2020 (Yamane & Kaneko, 2021). Females and those with higher education were less *pro-driving,* while those with a driver's license were more *pro-driving*. Homeowners were more *pro-driving* in spring 2020, while suburban or rural residents were more *pro-driving* in fall 2023. As expected, individuals from earlier birth cohorts were less *pro-driving*, likely due to physical limitations. The *pro-active* attitude was stronger among younger individuals in fall 2023, as well as among urban residents, college graduates, and those with higher incomes at both timepoints (Mondschein, 2021). Additionally, individuals with children and those without a driver's license were more *pro-active* in spring 2020, when most children were homeschooled, and alternative transportation such as walking and cycling became more prevalent than before the pandemic (Cusack, 2021), but this trend did not persist into fall 2023. Finally, younger individuals, those of Hispanic origin, individuals with higher household incomes, those with children in the household, and urban residents tend to be more *novelty-seeking*.

Table 4 Results of the structural model (N=1,612)

Variables	Categories	Pro-environment		Pro-driving		Pro-active		Novelty-seeking	
		In spring 2020	In fall 2023	In spring 2020	In fall 2023	In spring 2020	In fall 2023	In spring 2020	In fall 2023
Birth cohort (ref: Millennials or younger)	Generation X	-0.26 (-3.40)***			-0.07 (-0.61)		-0.36 (-2.60)***	-0.20 (-1.96)**	-0.10 (-1.01)
	Baby Boomers or older	-0.21 (-2.67)**			-0.20 (-1.98)**		-0.53 (-3.54)***	-0.60 (-6.15)***	-0.50 (-5.78)***
Gender (ref: non-female)	Female			-0.56 (-6.60)***	-0.63 (-7.39)***				
Ethnicity (ref: non-Hispanic)	Hispanic							0.42 (2.82)***	0.38 (2.80)***
Educational attainment (ref: lower than bachelor's degree)	Bachelor's degree or higher	0.63 (9.91)***	0.59 (10.62)***	-0.36 (-4.33)***	-0.56 (-6.93)***	0.61 (4.31)***	0.57 (4.17)***		
Household annual income (ref: less than $50,000)	$50,000 - $99,999					0.31 (1.85)*	0.43 (4.16)***	0.31 (3.20)***	0.22 (2.05)**
	$100,000 or more					0.72 (3.90)***	0.56 (3.90)***	0.52 (5.61)***	0.23 (2.67)***
Presence of children (ref: no)	Yes					0.50 (3.79)***		0.45 (4.47)***	0.35 (3.25)***
Housing Tenure (ref: rent or other)	Own			0.22 (2.54)**					
Neighborhood type (ref: urban)	Suburban	-0.33 (-4.68)***	-0.58 (-9.75)***		0.33 (3.53)**	-0.29 (-2.33)**	-0.40 (-2.90)***	-0.25 (-2.99)***	-0.21 (-2.56)***
	Rural	-0.06 (-0.61)	-0.42 (-4.17)***		0.38 (2.54)**	-0.05 (-0.24)	-0.31 (-1.49)	-0.44 (-3.05)***	-0.52 (-3.90)***
Having a driver's license (ref: no)	Yes			0.45 (4.21)***	0.67 (4.97)***	-0.79 (-4.62)***			

Note: Statistics in the table represent coefficients, robust t-statistics, significance level (*10%, **5%, ***1%).

5.3 Explanatory Variables Associated with Vehicle Transactions

Table 5 reports the results of the two multimodal logit models concurrently modeling past and expected future vehicle transactions with "*no transaction*" as the reference category. Between spring 2020 and fall 2023, individuals with a stronger *pro-environment* stance were less likely to have replaced vehicles than to have kept the same fleet, whereas those with greater proclivity for active lifestyles presented the opposite patterns. Additionally, individuals who are more *pro-driving* are more likely to expect to replace their vehicles in the future that to keep the same fleet. Not surprisingly, those with high scores on *novelty-seeking* have higher chances of vehicle transactions. They are more likely to acquire new vehicles, dispose of their current ones, or replace existing one(s) with new vehicles than to keep the same fleet.

Beyond indirect effects (through the latent attitudes), individuals' SDE traits also directly affect vehicle transactions. The youngest generations in the sample (Millennials and younger) were more likely to have increased their number of vehicles in the household compared to the oldest cohorts in the sample (Boomers and older). This trend is attributed to the dynamic nature of their household compositions, financial situations, lifestyles, and travel needs, all of which collectively contribute to a greater need for additional vehicles to accommodates commuting obligations and household responsibilities. A similar pattern emerges when comparing Millennials to Gen Xers regarding the expected vehicle transactions in the coming three years. Furthermore, females exhibited a lower inclination towards augmenting their vehicle counts. This aligns with existing research highlighting lower car dependence among women compared to men (den Braver et al., 2020; Guan & Wang, 2019). Additionally, within multi-member households, females may exert lesser influence over vehicle transaction decisions, especially if they are not employed. Households with children were more predisposed to have increased or to expect to increase their vehicle count.

We observe a positive correlation between the increase in the number of children/adults and the rise in household vehicles during the pandemic, consistent with prior research (Goodwill, 1993; Lee & Goulias, 2018; Thakuriah, 2024). Higher educational attainment in fall 2023 is associated with a higher likelihood of vehicle replacement in the subsequent years. In a similar vein, those with higher income levels were more likely to have increased their vehicle count in the past and to expect vehicle replacement in the future (Thakuriah, 2024). Households whose income has increased in the recent past were more likely to have added or replaced a vehicle(s) than those without income changes.

Holding all else constant, larger households tended to decrease or replace their cars from spring 2020 to fall 2023, compared to single-person households, and white individuals expect less increase in their vehicle ownership levels in the coming years. Given that larger or white households typically already own sufficient vehicles to meet their travel needs, they may have reduced their fleet size or have updated the fleet with a more efficient vehicle(s) during the pandemic.

Work status also plays a significant role in vehicle transactions. In fall 2023, those who expected to start or resume work in the next year (i.e., by fall 2024) were more likely to increase their vehicle count in the future, while those anticipating a pause or termination in employment were more likely to shed their vehicle(s). Regarding work arrangement, a higher monthly commuting frequency is associated with a decreased likelihood of vehicle reduction from spring 2020 to fall 2023, but an increased likelihood of vehicle addition or replacement of their vehicle in the future. Additionally, an increase in commuting frequency between spring 2020 and fall 2023 was negatively associated with vehicle-shedding during those years, while an expected

increase in remote working frequency after 2023 is positively associated with plans to replace a vehicle in the future.

Individuals who expressed concerns about the health impact of the pandemic in spring 2020 were less likely to have decreased their vehicle counts during that time. This trend reflects shifts towards private means of travel (e.g., driving) due to lingering health concerns during the pandemic (Loa et al., 2021; Zhang et al., 2021).

Finally, individuals who increased their household vehicle count during the pandemic are more likely to either increase, decrease or replace their vehicle count in the next three years. These individuals were able to add a vehicle during the pandemic (e.g., during a period with great financial challenge and uncertainty in society), and they appear to have continuous strong needs for vehicle transactions post-pandemic. Additionally, those who shed their vehicles are more inclined to acquire vehicles, while those who replaced vehicles are more likely to do so again in the future.

1 **Table 5 Results of the choice model (N=1,612)**

Variables	Categories	Past Vehicle Transaction (spring 2020 to fall 2023, "no transaction" as the reference)			Expected Future Vehicle Transaction (fall 2023 to fall 2026, "no transaction" as the reference)		
		Increase	**Decrease**	**Replace**	**Increase**	**Decrease**	**Replace**
Constants		-2.56	-2.11	-1.74	-1.98	-2.76	-2.13
		(-9.49)***	(-5.23)***	(-9.95)***	(-6.45)***	(-15.04)***	(-10.51)***
Attitudes							
Pro-environment				-0.14			
				(-2.09)**			
Pro-driving							0.24
							(3.06)***
Pro-active				0.24			
				(1.83)*			
Novelty-seeking					0.24	0.27	0.22
					(1.83)*	(1.76)*	(3.68)***
Socio-demographics and Neighborhood Characteristics							
Birth cohort	Generation X	-0.22			-0.50		
(ref[1]: Millennials or younger)		(-0.99)			(-2.24)**		
	Baby Boomers or older	-1.25			-0.58		
		(-4.17)***			(-1.99)**		
Gender	Female	-0.43					
(ref: non-female)		(-2.09)**					
Race	White only				-0.62		
(ref: not white only)					(-2.95)***		
Educational attainment	Bachelor's degree or higher						0.36
(ref: lower than bachelor's degree)							(2.52)**
Household size	Two members		1.31	0.68			0.63
(ref: one member)			(4.01)***	(3.50)***			(3.67)***
	Three or more members		1.26	1.02			0.67
			(3.71)***	(5.12)***			(3.65)***
Presence of a child(ren)	Yes	1.07			1.29		
(ref: no)		(4.98)***			(5.82)***		
Annual household income	$50,000 - $99,999	0.39					0.75
(ref: less than $50,000)		(1.26)					(3.91)***
	$100,000 or more	0.91					0.71
		(2.96)***					(3.70)***
Work Arrangement before the Pandemic (in 2019, retrospectively reported in spring 2020) vs. fall 2023							
Monthly commute days			-0.04		0.02		0.01
			(-2.96)***		(1.77)*		(2.11)**
COVID Health Concerns (only Measured in Spring 2020)							
Level of concerns	Somewhat concerned		-0.93		---	---	---
(ref: not concerned or neutral)			(-2.79)***		---	---	---
	Strongly concerned		-0.93		---	---	---
			(-3.09)***		---	---	---
Life Events and Changes in Work Arrangements							
Changes in Employment status	Worker → non-worker				0.24	1.25	

Variables	Categories	Past Vehicle Transaction (spring 2020 to fall 2023, "no transaction" as the reference)			Expected Future Vehicle Transaction (fall 2023 to fall 2026, "no transaction" as the reference)		
		Increase	**Decrease**	**Replace**	**Increase**	**Decrease**	**Replace**
(ref: no change)					(0.32)	(2.20)**	
	Non-worker → worker				1.23	-0.67	
					(3.27)***	(-0.65)	
Changes in Household income	Increased	0.84		0.38	---	---	---
(ref: no change)		(3.75)***		(2.78)***	---	---	---
Changes in Number of kids (<18)	Increased	0.74		0.69	---	---	---
(ref: no change)		(2.05)**		(2.70)***	---	---	---
Changes in Number of adults (18-64)	Increased	0.80		1.05	---	---	---
(ref: no change)		(3.06)***		(3.81)***	---	---	---
Change in commute days	Increased/Expected to increase		-0.04	-0.02			
			(-2.75)***	(-2.51)**			
Change in remote working days	Increased/Expected to increase						0.02
							(2.31)**
Past Household Vehicle Count Changes (from spring 2020 to fall 2023)							
Changes in vehicle count	Increased	---	---	---	1.05	1.39	0.45
(ref: no change)		---	---	---	(3.27)***	(3.87)***	(1.86)*
	Decreased	---	---	---	1.35		
		---	---	---	(4.73)***		
	Replaced	---	---	---			0.69
		---	---	---			(5.24)***
# of Observation		Increase 117; Decrease 119; Replace 381; No transaction 995			Increase 126; Decrease 67; Replace 529; No transaction 890		
LL(0)				-2137.67			-2137.67
LL(final model)				-1486.68			-1449.06

Notes:
1. "ref" represents the reference category for a given categorical variable.
2. Statistics in the table represent coefficients, robust t-statistics, and significance level (*10%, **5%, ***1%).
3. Blank cells indicate variables that are tested in the model but not statistically significant (and removed from the final specification), , whereas cells with "---" indicate variables that are not tested in the model due to lack of information.

6 Discussions

The study made contributions in several ways. First, it adopted a longitudinal approach to examine changes in vehicle ownership level. Second, it classified changes in vehicle ownership level into four categories, namely *increase*, *decrease*, *replace*, and *no transaction*. By doing so, it considered both the total "number" of the vehicles and presence of changes to the vehicle fleet maintaining the same total. Third, the study explored the impacts of less-studied factors on vehicle transactions, including individual attitudes, life events, work arrangements and COVID health concerns. Finally, by embedding latent attitudes in an ICLV model, the study explicitly controls for key dimensions of individual-level heterogeneity, such as *pro-driving* and *novelty-seeking*, that are known to influence vehicle-related decisions. Importantly, these latent attitudes are measured at two timepoints, allowing them to capture stable preferences that influence both past and future vehicle transactions. Even after accounting for these latent traits, we still observe strong and statistically significant effects of past vehicle transactions on expected future transactions. This provides compelling evidence of genuine behavioral state dependence.

The study found that individuals who are *pro-driving* and *novelty-seeking* are more likely to consider increasing or replacing their vehicles in the future. This is consistent with other studies showing that individuals with these attitudes are also more inclined to embrace new vehicle technologies, such as electric vehicles (Iogansen et al., 2023), autonomous vehicles (Wang & Akar, 2019), and shared mobility services, such as ridehailing (Lavieri & Bhat, 2019), carsharing (Mueller et al., 2015) and micromobility (Mahmoud et al., 2021). Policies aimed at incentivizing these individuals to transition to cleaner vehicles or promoting mode shift away from private vehicles altogether, could have significant impacts on reducing carbon emissions. For instance, public agencies may consider pivoting from broad-based incentives to targeted "Early Adopter Grants" for Mobility-as-a-Service (MaaS) subscriptions, specifically marketing these as "high-tech" and "cutting-edge" lifestyle choices. MaaS providers could offer tiered "beta-tester" memberships that provide exclusive access to new vehicle types or autonomous shuttle pilots.

Individuals from younger generations have higher likelihood in increasing vehicle count compared to their older counterparts, likely due to their more dynamic household composition, financial conditions, and student/work status. Also, higher educational attainment and income levels are positively associated with vehicle addition and replacement. It is important to formulate policies that divert these segments of individuals away from increasing vehicle ownership level, especially among those who are currently non-vehicle owners. For instance, public agencies may partner with employers to offer "New Hire Mobility Wallets". Instead of traditional parking subsidies, companies provide a pre-tax monthly credit usable exclusively for MaaS, public transit, or e-bike subscriptions. This targets young professionals at a critical "habit-breaking" window (i.e., the start of a new job), preventing the "first car" purchase habit from forming.

The presence of children is positively associated with increased vehicle counts, and an increase in the number of kids and adults in the household is linked to an increase in vehicle counts. A deeper understanding of how vehicles are shared and used among household numbers, and how daily trip chaining patterns are structured would be crucial to help policymakers and car manufacturers develop transportation policies and designing vehicles that meet the diverse needs of larger families.

In terms of work status, those who expected a transition to employment in the near future also expect to increase their number of vehicles in the household. Commuting is also positively associated with a lower likelihood of shedding vehicles during the pandemic and a higher likelihood of purchasing/replacing vehicles in the future. However, we did not observe significant impacts of remote work on vehicle count in the past. As commuter vehicles are often retained for other household activities, cities may need to shift from commuter-oriented transit hubs toward neighborhood-based mobility hubs that are integrated into residential areas. Such a transition could help disrupt habitual car use for non-work trips, such as errands and shopping, that persists even when commuting is reduced or eliminated.

While the ICLV model offers valuable insights into the impact of exogenous variables on the outcomes, model coefficients alone may not clearly reflect the magnitude of these effects and their policy implications. Accordingly, we use the Average Treatment Effect (ATE) (Heckman & Vytlacil, 2000) to quantify the impact of specific variables by assessing how the average predicted probability of each outcome changes when a variable transitions from one state to another, while holding all other factors constant. For example, if altering work arrangements (e.g., reducing commute frequency or increasing remote work frequency) is found to reduce vehicle count, then implementing commute trip reduction policies and remote working policies, would lead to a reduction in vehicle ownership level. To estimate the ATE of work arrangements, workers are assumed to be in four distinct states (S1- S4) relative to the base state (S0), in term of the level of commuting frequency. Then, using the estimated ICLV model, we can compute the probability of each alternative within the choice set for each state (S0-S4) at the individual level. The ATE is the difference between the average predicted probability for state S1- S4 and S0. The states are defined as follows:

- S0: self-reported monthly commute frequency;
- S1: commute 20 days per month (i.e., fully commute as the lower bound of commute reduction intervention);
- S2: reduce the self-reported monthly commute frequency by 25% (weaker intervention);
- S3: reduce the self-reported monthly commute frequency by 50% (stronger intervention);
- S4: commute zero days per month (i.e., fully remote work as the upper bound of intervention).

Table 6 presents *changes* in the average predicted probability of past and expected future vehicle transactions (in percentage points or p.p.) using the ATE for commuting frequency. Reducing commute trips during the pandemic led to a greater increase in probability of vehicle-shedding, compared to post-pandemic period (e.g., a 3.16 p.p. increase in the past versus a 0.31 p.p. increase in the future). Not surprisingly, the effect is larger during the pandemic period, at a time in which travel was largely restricted: A lower number of commuting trips would lead to household vehicles being used less, thereby encouraging vehicle shedding. In contrast, by fall 2023, the demand for non-commuting trips had largely rebounded. Even if workers reduced their number of commute trips, household vehicles would be still likely needed for other purposes, reducing the probability of vehicle shedding. Nevertheless, it is still expected that reducing commute trips will decrease the likelihood of increasing household vehicles in the future (e.g., a 1.11 p.p. decrease by individuals with zero commutes).

Table 6 Average treatment effect of work arrangements (percentage-point differences from the baseline estimates)

State	Interventions in work arrangement status	Past changes (among those who were workers in spring 2020 and fall 2023, n=799)			
		Increase	Decease	Replace	No change
S1	Fully commute (commute 20 days/month)	0.43	-1.70	-1.04	2.30
S2	Reducing a quarter of current commute frequency	0.00	0.63	-0.47	-0.15
S3	Reducing half of current commute frequency	-0.02	1.35	-0.96	-0.38
S4	No commute (commute 0 days/month)	-0.09	3.16	-1.98	-1.09

State	Interventions in work arrangement status	Expected changes (among those who were workers in fall 2023 and expected to be workers in fall 2024, n=799)			
		Increase	Decease	Replace	No change
S1	Fully commute (commute 20 days/month)	0.70	-0.21	1.78	-2.27
S2	Reducing a quarter of current commute frequency	-0.29	0.08	-0.64	0.85
S3	Reducing half of current commute frequency	-0.57	0.16	-1.29	1.71
S4	No commute (commute 0 days/month)	-1.11	0.31	-2.59	3.39

Notes:

- State S1 to S4 were established by updating the values in two variables, "monthly commute days" and "change in monthly commute days".
- Respondents who already commute more than 20 days/month maintain their current commute frequency in the S1 state.
- The values in the table are changes in the average predicted probability of past and expected changes in vehicle count (in percentage points).

If governments have the goal of promoting reduction in vehicle ownership (and use), they could consider supporting company polices aiming at less-frequent commute trips and promoting remote and hybrid work arrangements. However, the impact of such policies can vary depending on the level of flexibility offered. For instance, partial-day hybrid work seems to offset the trip reduction benefit from full-day remote working. Policies also need to be customized for different population segments: Individuals who have the option to work remotely entirely will likely reduce the number of vehicles they own. However, those who still need to commute might still need good access to vehicles to access jobs and opportunities.

The study sheds light on the relationship of a wide range of factors and the type of vehicle transactions, which may not necessarily correspond to the impact of the same factors on vehicle ownership (or counts). Note that the number of household vehicles and its changes during a certain period are two interrelated, but distinct choices, each of which could be driven by unique decision-making processes. For instance, being *pro-driving* is found not to directly affect the changes in vehicle count during the pandemic; however, in academic literature, a positive association of *pro-driving* attitudes and vehicle ownership and usage are well established.

Finally, consistent with Kitamura & Bunch (1990), the findings reveal a noticeable behavioral asymmetry regarding specific factors. For instance, an increase in income and the number of adults and children in the household is associated with a significantly higher likelihood of vehicle augmentation or replacement. Conversely, a decrease in income and the number of adults and children does not appear to have a corresponding effect on reducing vehicle counts, once household habits for car use have been established. This suggests a potential

"ratchet effect" where households become accustomed to a certain level of mobility and vehicle ownership, reinforcing persistent travel habits even when underlying needs change. To address this inertia, policymakers may consider implementing “Life-Stage Transition Incentives”. For example, agencies could offer temporary "downsizing incentives", such as property tax credits or subsidized public transit passes, to households that voluntarily reduce fleet size following qualifying life events. Such measures could help counteract the tendency to retain underutilized vehicles and encourage more durable adjustments in travel behavior.

7 Conclusions

Using a panel survey dataset comprising 1,612 U.S. residents, this study explored the factors influencing household vehicle transactions retrospectively between spring 2020 and fall 2023 and prospectively between fall 2023 and fall 2026. We employed an ICLV model that jointly estimated two multinomial logit models, with latent attitudes, SDE characteristics, life events, work arrangements and COVID-related health concerns as explanatory variables. The results indicate that *pro-driving*, *pro-environment,* and *pro-active* individuals were more likely to replace their vehicles, while *novelty-seeking* individuals exhibited a higher propensity to increase, decrease, and replace their vehicles. Moreover, younger individuals, households with children, and those with an increase in the number of children or adults in the household had a higher likelihood of acquiring vehicles, likely to accommodate their dynamic travel needs. Transitions into the workforce and an increase in household income during the study period were associated with an increased likelihood of vehicle acquisition. Furthermore, an increase in commute frequency was associated with a lower likelihood of vehicle shedding in the past, and higher likelihood of vehicle purchasing in the future. Not surprisingly, COVID health concerns led to a lower chance of vehicle shedding during the pandemic. Finally, households that increased vehicle count during the pandemic are likely to adjust it again, while those who shed vehicles tend to reacquire them, and those who replaced vehicles are inclined to do so again. Planners and policymakers could leverage the findings on the impacts of latent attitudes, SED characteristics and life events on vehicle transactions to cater to individuals and households with diverse socioeconomic backgrounds. Employers could lead households to owning fewer vehicles by promoting remote and hybrid work arrangements.

This study presents several limitations. First, survey respondents were asked to report only one type of change by comparing vehicle counts at specific timepoints. However, respondents may have experienced or expected “multiple” changes between these time points. For instance, an individual may have replaced one vehicle and subsequently shed another. Unfortunately, in a case like this, the survey data only captures the behavior of vehicle shedding, neglecting the vehicle replacement. Using mutually exclusive choices, such as *replacement*, *disposal without replacement*, and *acquisition without disposal*, used by Yamamoto (2008), can resolve this issue. Second, the choice outcomes provided by respondents solely indicate the direction of change in vehicle count, rather than the magnitude: If a respondent reported a decrease in their household vehicle count, we cannot discern whether vehicle count decreases by one vehicle or more. Third, despite testing a wide range of model specifications, certain key household-level attributes cannot be controlled in the study. Vehicle ownership tend to be a household decision (Woldeamanuel et al., 2009). For example, fluctuations in the number of household workers could impact on the changes in vehicle counts, but this information was not collected in the survey. Moreover, although attitudes are modeled at two timepoints, it remains

possible that vehicle ownership experiences between survey waves influenced respondents' attitudinal orientations. Future studies could further disentangle the bidirectional relationship between attitudes and vehicle transactions. Finally, although the study leverages a rich longitudinal dataset with detailed attitudinal and behavioral information, the sample is not nationally representative. As a result, estimated effect sizes and predicted probabilities should not be interpreted as population-level forecasts. Instead, the findings are best understood as identifying behavioral mechanisms, relative sensitivities, and sources of heterogeneity in vehicle transaction decisions. Policy implications derived from these results should therefore be viewed as illustrative and hypothesis-generating, rather than prescriptive estimates of aggregate impacts.

We propose several avenues for future research. First, given the rapid evolution of the auto industry and surge in vehicle transactions as the economy recovered, continued efforts to study this topic could yield critical insights in the years ahead. This study can be expanded by investigating vehicle fleet composition and fuel type choices. Research questions related to which vehicles are the first to be discarded during income declines, or whether compact cars are replaced with larger crossovers or SUVs in response to increased household size are worthy of investigation. The data available for this study include details of all passenger vehicles available to the households (including make, model, year, and fuel type) in spring 2020 and fall 2023. Leveraging this information could allow for deeper insights into the ways that households upgrade or downgrade (or upsize or downsize) their vehicles. Third, our model did not detect the influence of neighborhood characteristics on changes in vehicle count. Introducing location-based variables into the analysis may provide additional useful insights, for example, when studying transit-rich vs. transit-poor neighborhoods/cities. Further exploration of this phenomenon can shed more light on the interplay between neighborhood characteristics and vehicle ownership dynamics. Finally, statistical associations in this study do not necessarily indicate causal relationships among variables, and more research is called for to establish causality.

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